

Project Team Task Assignment & Responsibility Plan

Project Phase

Pre-Implementation Phase

Team Leader

Mustafa Mahmoud

Team Members

Mustafa Mahmoud

Mina Safwat Fawzy

Omar Moharam

Ammar Amr

Mohamed Saber

Document Purpose

This document defines task ownership, responsibilities, deliverables, timelines, dependencies and completion criteria for the pre-implementation phase of the EgySpy project (04/10/2026 to 30/10/2026). It is the team's practical reference for who does what, by when, with whose help, and how to confirm that each task is finished correctly.

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2. Team Responsibility Overview

Rank	Member	Role	General Responsibility
1	Mustafa Mahmoud	Team Leader	Critical tasks, architecture, major decisions, reviews
2	Mina Safwat Fawzy	Team Member	Important analysis/design responsibilities
3	Omar Moharam	Team Member	Analysis/design/documentation
4	Ammar Amr	Team Member	Analysis/design/documentation
5	Mohamed Saber	Team Member	Analysis/design/documentation

Reading notes: dates use DD/MM/YYYY and are taken unchanged from the Gantt Chart. Duration is counted in calendar days, including the start and end dates. Tasks are numbered in chronological order of start date (Gantt order is kept for tasks starting on the same day). Task names are the Gantt Chart names.

3. Master Task Assignment Table

#	Task	Phase	Primary Owner	Supporting Members	Start Date	End Date	Duration	Priority	Complexity	Main Deliverable
1	Project proposal consolidation	Project Planning & Management	Mustafa Mahmoud	Mohamed Saber	04/10/2026	07/10/2026	4 days	High	Medium	Consolidated proposal (baseline v1)
2	Review scope, research questions & source-collection plan	Literature Review	Mohamed Saber	None	04/10/2026	06/10/2026	3 days	Medium	Low	Literature review plan
3	Stakeholder analysis	Requirements Gathering	Omar Moharam	None	05/10/2026	08/10/2026	4 days	Medium	Low	Stakeholder register and matrix
4	Review of UWB/RTLS workforce tracking and smart attendance systems	Literature Review	Mohamed Saber	None	06/10/2026	12/10/2026	7 days	Medium	Medium	UWB/RTLS and attendance systems review section
5	Review of anti-tampering / anti-fraud mechanisms	Literature Review	Mina Safwat Fawzy	None	07/10/2026	13/10/2026	7 days	High	High	Anti-tampering review section with threat/countermeasure table
6	Task Assignment & Roles Planning	Project Planning & Management	Mustafa Mahmoud	None	08/10/2026	11/10/2026	4 days	High	Low	Approved task assignment and responsibility plan
7	Review of real-time tracking & streaming technologies	Literature Review	Omar Moharam	None	08/10/2026	14/10/2026	7 days	Medium	Medium	Real-time tracking and streaming review section
8	User stories	Requirements Gathering	Ammar Amr	None	08/10/2026	12/10/2026	5 days	High	Medium	Prioritized user stories with acceptance criteria
9	Problem statement, objectives & scope refinement	System Analysis & Design	Mustafa Mahmoud	None	08/10/2026	11/10/2026	4 days	High	Medium	Problem statement, objectives and scope statement
10	Risk assessment & mitigation plan	Project Planning & Management	Mohamed Saber	None	09/10/2026	14/10/2026	6 days	Medium	Medium	Risk register with mitigation plan
11	Functional requirements	Requirements Gathering	Ammar Amr	None	09/10/2026	15/10/2026	7 days	Critical	High	Functional requirements list
12	Use cases	Requirements Gathering	Omar Moharam	None	10/10/2026	14/10/2026	5 days	High	Medium	Use case list with actors and goals
13	Weekly progress reviews & plan updates	Project Planning & Management	Mustafa Mahmoud	None	11/10/2026	29/10/2026	19 days	High	Low	Weekly status notes and updated plan
14	Non-functional requirements	Requirements Gathering	Mina Safwat Fawzy	None	11/10/2026	16/10/2026	6 days	Critical	High	Non-functional requirements list with measurable targets
15	KPIs / project success metrics	Project Planning & Management	Mohamed Saber	None	12/10/2026	16/10/2026	5 days	Medium	Low	KPI list with targets
16	Architecture analysis & candidate evaluation	System Analysis & Design	Mina Safwat Fawzy	Mustafa Mahmoud Omar Moharam	12/10/2026	16/10/2026	5 days	Critical	High	Candidate comparison matrix with recommendation
17	Use case diagram	System Analysis & Design	Omar Moharam	None	13/10/2026	16/10/2026	4 days	High	Medium	Use case diagram
18	Comparative synthesis & gap analysis	Literature Review	Mohamed Saber	None	14/10/2026	17/10/2026	4 days	Medium	Medium	Comparison table and gap list
19	Literature review document drafting	Literature Review	Mohamed Saber	Mina Safwat Fawzy Omar Moharam	15/10/2026	19/10/2026	5 days	Medium	Medium	Draft literature review document
20	Use case descriptions	System Analysis & Design	Omar Moharam	None	15/10/2026	19/10/2026	5 days	High	Medium	Use case description set
21	Requirements analysis	System Analysis & Design	Mina Safwat Fawzy	None	15/10/2026	19/10/2026	5 days	Critical	High	Requirements analysis report and priority matrix

#	Task	Phase	Primary Owner	Supporting Members	Start Date	End Date	Duration	Priority	Complexity	Main Deliverable
22	Requirements validation / review	Requirements Gathering	Mustafa Mahmoud	Mohamed Saber Omar Moharam	16/10/2026	18/10/2026	3 days	High	Medium	Validation report with issue log
23	Technology stack evaluation & selection	System Analysis & Design	Mustafa Mahmoud	Ammar Amr	16/10/2026	20/10/2026	5 days	Critical	High	Evaluation matrix and selected stack
24	Requirements documentation (baseline SRS)	Requirements Gathering	Ammar Amr	None	17/10/2026	20/10/2026	4 days	Critical	Medium	Baseline SRS document
25	Architecture selection & high-level architecture	System Analysis & Design	Mustafa Mahmoud	Mina Safwat Fawzy Ammar Amr	17/10/2026	20/10/2026	4 days	Critical	High	High-level architecture diagram and decision record
26	ERD / conceptual data model	System Analysis & Design	Mina Safwat Fawzy	Ammar Amr	17/10/2026	20/10/2026	4 days	Critical	High	Conceptual ERD with entity descriptions
27	UI/UX guidelines	System Analysis & Design	Omar Moharam	None	17/10/2026	20/10/2026	4 days	Medium	Low	UI/UX guideline document
28	Context-level DFD	System Analysis & Design	Mohamed Saber	None	18/10/2026	19/10/2026	2 days	Medium	Low	Context-level DFD
29	Wireframes	System Analysis & Design	Omar Moharam	None	18/10/2026	22/10/2026	5 days	Medium	Medium	Wireframe set
30	Major components & their interactions	System Analysis & Design	Mustafa Mahmoud	Omar Moharam	19/10/2026	22/10/2026	4 days	Critical	High	Component list and interaction overview
31	Detailed DFD	System Analysis & Design	Mohamed Saber	None	19/10/2026	22/10/2026	4 days	Medium	Medium	Detailed DFD
32	Feedback & evaluation	Literature Review	Mohamed Saber	None	20/10/2026	22/10/2026	3 days	Low	Low	Feedback log with priorities
33	Requirements consistency review	System Analysis & Design	Mohamed Saber	None	20/10/2026	21/10/2026	2 days	High	Medium	Consistency review log
34	Logical schema	System Analysis & Design	Mina Safwat Fawzy	None	20/10/2026	23/10/2026	4 days	Critical	High	Logical schema document
35	Sequence diagrams	System Analysis & Design	Omar Moharam	None	20/10/2026	24/10/2026	5 days	High	Medium	Sequence diagram set
36	Integration considerations	System Analysis & Design	Ammar Amr	None	20/10/2026	23/10/2026	4 days	Medium	Medium	Integration considerations document
37	Activity & state diagrams	System Analysis & Design	Ammar Amr	None	21/10/2026	24/10/2026	4 days	Medium	Medium	Activity and state diagrams
38	Component diagram	System Analysis & Design	Mohamed Saber	None	21/10/2026	24/10/2026	4 days	Medium	Medium	Component diagram
39	Suggested improvements & revisions	Literature Review	Mohamed Saber	None	22/10/2026	24/10/2026	3 days	Low	Low	Revised literature review and list of applied changes
40	Physical schema	System Analysis & Design	Mina Safwat Fawzy	None	22/10/2026	25/10/2026	4 days	High	High	Physical schema document
41	Mockups / interactive prototype	System Analysis & Design	Omar Moharam	Ammar Amr	22/10/2026	27/10/2026	6 days	Medium	High	Mockups and clickable prototype
42	Deployment diagram	System Analysis & Design	Ammar Amr	None	22/10/2026	25/10/2026	4 days	Medium	Medium	Deployment diagram
43	Class diagram	System Analysis & Design	Ammar Amr	None	23/10/2026	26/10/2026	4 days	Medium	Medium	Class diagram
44	Testing & validation planning	System Analysis & Design	Ammar Amr	Mohamed Saber	23/10/2026	27/10/2026	5 days	Medium	Medium	Test and validation plan

#	Task	Phase	Primary Owner	Supporting Members	Start Date	End Date	Duration	Priority	Complexity	Main Deliverable
45	API documentation	System Analysis & Design	Mina Safwat Fawzy	Omar Moharam	24/10/2026	27/10/2026	4 days	High	High	API specification document
46	Final review against grading/evaluation criteria	Literature Review	Mohamed Saber	None	25/10/2026	27/10/2026	3 days	Medium	Low	Criteria checklist and final literature review
47	Deployment strategy	System Analysis & Design	Ammar Amr	None	25/10/2026	27/10/2026	3 days	Medium	Low	Deployment strategy document
48	Planning documents finalization	Project Planning & Management	Ammar Amr	None	26/10/2026	30/10/2026	5 days	Medium	Low	Final planning documents package
49	Documentation consolidation	System Analysis & Design	Mina Safwat Fawzy	Mohamed Saber Ammar Amr	26/10/2026	30/10/2026	5 days	High	Medium	Consolidated document set
50	Cross-deliverable integration review & corrections	System Analysis & Design	Mustafa Mahmoud	Mina Safwat Fawzy Omar Moharam Ammar Amr Mohamed Saber	27/10/2026	29/10/2026	3 days	Critical	High	Integration review log and confirmed corrections
51	Final system design review	System Analysis & Design	Mustafa Mahmoud	None	28/10/2026	29/10/2026	2 days	Critical	Medium	Final review sign-off
52	Final consistency check & deliverable preparation	System Analysis & Design	Mustafa Mahmoud	None	30/10/2026	30/10/2026	1 day	Critical	Medium	Final deliverable package

4. Timeline Overview

Phase	Start	End	Main Activities
Project Planning & Management	04/10/2026	30/10/2026	Proposal consolidation, task assignment and roles, risk assessment, KPIs, weekly progress reviews, planning document finalization
Literature Review	04/10/2026	27/10/2026	Review scope, three focused reviews (UWB/RTLS, anti-tampering, real-time streaming), synthesis and gap analysis, drafting, feedback, revisions, final review
Requirements Gathering	05/10/2026	20/10/2026	Stakeholder analysis, user stories, use cases, functional and non-functional requirements, validation, baseline SRS
System Analysis & Design	08/10/2026	30/10/2026	Scope refinement, use case modeling, requirements analysis, architecture, data design, process and behavior modeling, UI/UX, technology, API, testing and deployment planning, integration review, documentation consolidation, final checks

Stages inside System Analysis & Design (grouping of Gantt tasks)

Stage	Start	End	Tasks
Analysis	08/10/2026	21/10/2026	9 Problem statement, objectives & scope refinement; 17 Use case diagram; 20 Use case descriptions; 21 Requirements analysis; 33 Requirements consistency review
Architecture	12/10/2026	22/10/2026	16 Architecture analysis & candidate evaluation; 25 Architecture selection & high-level architecture; 30 Major components & their interactions
Data & process modeling	17/10/2026	25/10/2026	26 ERD / conceptual data model; 28 Context-level DFD; 31 Detailed DFD; 34 Logical schema; 40 Physical schema
Behavior & structure modeling	20/10/2026	26/10/2026	35 Sequence diagrams; 37 Activity & state diagrams; 38 Component diagram; 42 Deployment diagram; 43 Class diagram
UI/UX	17/10/2026	27/10/2026	27 UI/UX guidelines; 29 Wireframes; 41 Mockups / interactive prototype
Technical planning	16/10/2026	27/10/2026	23 Technology stack evaluation & selection; 36 Integration considerations; 44 Testing & validation planning; 45 API documentation; 47 Deployment strategy
Finalization	26/10/2026	30/10/2026	49 Documentation consolidation; 50 Cross-deliverable integration review & corrections; 51 Final system design review; 52 Final consistency check & deliverable preparation

Overall flow: Literature Review and Requirements Gathering run first (04/10 to 20/10) and feed System Analysis & Design, whose modeling work peaks between 17/10 and 27/10. The last three days (27/10 to 30/10) are reserved for integration review, final design review, documentation consolidation and the final consistency check.

5. Individual Task Specifications

Task 1 - Project proposal consolidation

5.1 Basic Information

Attribute	Details
Phase	Project Planning & Management
Primary Owner	Mustafa Mahmoud
Supporting Members	Mohamed Saber
Start Date	04/10/2026
End Date	07/10/2026
Duration	4 days
Priority	High
Complexity	Medium
Dependencies	None

5.2 Task Objective

Combine the August proposal and all team input into one agreed description of the project's scope, objectives and approach. Every later planning, requirements and design task uses it as the common reference. It prevents team members from working from different understandings.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Decide the final wording of scope, objectives and approach
- Resolve conflicting inputs from members
- Get the team's agreement on the final text

Supporting Member - Mohamed Saber

- Compile team inputs and edits into one draft
- Keep version control and track open comments

5.4 Expected Deliverables

- Consolidated proposal (baseline v1): background, objectives, scope boundaries, high-level approach and assumptions

5.5 Definition of Done

- All five members have confirmed the content
- Scope and objectives are stated once, with no conflicting versions
- The text states that this period covers planning and design only, not implementation

5.6 Key Quality Checks

- ☐ Content matches the August proposal except where deliberately changed
- ☐ Architecture and technology are treated as open, not locked
- ☐ Scope boundaries are explicit

5.7 Handoff / Dependency

Output Used By: Task 3 - Stakeholder analysis; Task 6 - Task Assignment & Roles Planning; Task 9 - Problem statement, objectives & scope refinement; Task 10 - Risk assessment & mitigation plan; Task 15 - KPIs / project success metrics

Handoff: Consolidated proposal to Mustafa (Problem statement), Omar (Stakeholder analysis) and Mohamed (Review scope) as the shared reference.

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check that scope and objectives are clear, consistent and detailed enough to base requirements on.

Task 2 - Review scope, research questions & source-collection plan

5.1 Basic Information

Attribute	Details
Phase	Literature Review
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	04/10/2026
End Date	06/10/2026
Duration	3 days
Priority	Medium
Complexity	Low
Dependencies	None

5.2 Task Objective

Define what the literature review covers, which questions it must answer and where sources come from. It keeps the three review topics focused and avoids overlap between authors.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Write the scope and 3 to 5 research questions
- Define source types and search keywords
- Assign each review topic to an author

5.4 Expected Deliverables

- Review plan: scope, research questions, source approach and inclusion criteria

5.5 Definition of Done

- Scope and questions are agreed by the team
- The source collection method is clear
- Each review topic has a named author

5.6 Key Quality Checks

- ☐ Questions match the project objectives
- ☐ UWB/attendance, anti-tampering and streaming topics are all covered
- ☐ The plan is realistic for the time available

5.7 Handoff / Dependency

Output Used By: Task 4 - Review of UWB/RTLS workforce tracking and smart attendance systems; Task 5 - Review of anti-tampering / anti-fraud mechanisms; Task 7 - Review of real-time tracking & streaming technologies

Handoff: Review plan to Mohamed (UWB/RTLS review), Mina (anti-tampering review) and Omar (streaming review).

5.8 Review / Approval

Reviewed By: Omar Moharam

Review Responsibility: Check that the scope is clear and covers all required topics.

Task 3 - Stakeholder analysis

5.1 Basic Information

Attribute	Details
Phase	Requirements Gathering
Primary Owner	Omar Moharam
Supporting Members	None
Start Date	05/10/2026
End Date	08/10/2026
Duration	4 days
Priority	Medium
Complexity	Low
Dependencies	Task 1 - Project proposal consolidation

5.2 Task Objective

Identify everyone who uses, manages or is affected by the system and describe their needs. Requirements are incomplete if a stakeholder group is missed.

5.3 Task Responsibilities

Primary Owner - Omar Moharam

- Identify all stakeholder groups and their roles
- Describe each group's needs, expectations and influence
- Prepare a stakeholder matrix

5.4 Expected Deliverables

- Stakeholder register with a needs description and an influence/interest matrix

5.5 Definition of Done

- All relevant groups are listed
- Needs are described for each group
- The matrix is approved by the team

5.6 Key Quality Checks

- ☐ No major group is missing (for example employee, manager, administrator, HR)
- ☐ Needs are specific
- ☐ Content matches the proposal scope

5.7 Handoff / Dependency

Output Used By: Task 8 - User stories; Task 12 - Use cases; Task 14 - Non-functional requirements

Handoff: Stakeholder register to Ammar (User stories) and Omar (Use cases).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check coverage of stakeholder groups and clarity of their needs.

Task 4 - Review of UWB/RTLS workforce tracking and smart attendance systems

5.1 Basic Information

Attribute	Details
Phase	Literature Review
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	06/10/2026
End Date	12/10/2026
Duration	7 days
Priority	Medium
Complexity	Medium
Dependencies	Task 2 - Review scope, research questions & source-collection plan

5.2 Task Objective

Survey existing UWB, RTLS and smart attendance systems to understand current solutions, their accuracy and their limits. This gives the team a factual base for design decisions and the gap analysis.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Collect and read relevant papers, products and case studies
- Summarize each system: technology, accuracy, strengths and limits
- Write the review section with references

5.4 Expected Deliverables

- Review section with a comparison table of surveyed systems and references

5.5 Definition of Done

- Enough relevant sources are covered
- Each source is summarized in the agreed format
- References are complete

5.6 Key Quality Checks

- ☐ Sources are credible and recent where possible
- ☐ Limitations of each system are stated
- ☐ Findings are written in the team's own words with citations

5.7 Handoff / Dependency

Output Used By: Task 16 - Architecture analysis & candidate evaluation; Task 18 - Comparative synthesis & gap analysis

Handoff: Review section to Mohamed (Comparative synthesis) and Mina (Architecture analysis, as technology input).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check coverage, accuracy of summaries and quality of sources.

Task 5 - Review of anti-tampering / anti-fraud mechanisms

5.1 Basic Information

Attribute	Details
Phase	Literature Review
Primary Owner	Mina Safwat Fawzy
Supporting Members	None
Start Date	07/10/2026
End Date	13/10/2026
Duration	7 days
Priority	High
Complexity	High
Dependencies	Task 2 - Review scope, research questions & source-collection plan

5.2 Task Objective

Study how attendance and tracking systems are attacked (spoofing, relay, device tampering, buddy punching) and which countermeasures exist. Anti-tampering is a core feature of the project, so this review feeds the requirements and the architecture.

5.3 Task Responsibilities

Primary Owner - Mina Safwat Fawzy

- Collect sources on attacks and defenses relevant to UWB tracking
- Build a threat and countermeasure table
- Write the review section and mark the points that affect requirements

5.4 Expected Deliverables

- Review section with a threat/countermeasure table

5.5 Definition of Done

- Main threat types are covered
- Each threat has at least one documented countermeasure from sources
- Findings relevant to security requirements are marked

5.6 Key Quality Checks

- ☐ Threats are relevant to workforce attendance
- ☐ Countermeasures are realistic for the project scope
- ☐ Sources are cited

5.7 Handoff / Dependency

Output Used By: Task 14 - Non-functional requirements; Task 16 - Architecture analysis & candidate evaluation; Task 18 - Comparative synthesis & gap analysis

Handoff: Findings to Mina's Non-functional requirements and Architecture analysis, and to Mohamed for the synthesis.

5.8 Review / Approval

Reviewed By: Omar Moharam

Review Responsibility: Check that threats are relevant and the countermeasures are supported by sources.

Task 6 - Task Assignment & Roles Planning

5.1 Basic Information

Attribute	Details
Phase	Project Planning & Management
Primary Owner	Mustafa Mahmoud
Supporting Members	None
Start Date	08/10/2026
End Date	11/10/2026
Duration	4 days
Priority	High
Complexity	Low
Dependencies	Task 1 - Project proposal consolidation

5.2 Task Objective

Fix who owns, supports and reviews every task in the Gantt Chart. Clear ownership avoids duplicated or forgotten work. This document is the main output of the task.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Finalize owners, supporting members and reviewers
- Define handoffs between dependent tasks
- Present the plan to the team and record confirmation

5.4 Expected Deliverables

- Approved assignment plan: owner, supporting members and reviewer for each task

5.5 Definition of Done

- Every Gantt task has exactly one Primary Owner
- Every member has confirmed their tasks
- Handoffs between dependent tasks are written down

5.6 Key Quality Checks

- ☐ No task is missing or duplicated
- ☐ Gantt dates are unchanged
- ☐ No member has an unrealistic overlap of high-effort tasks

5.7 Handoff / Dependency

Output Used By: Task 13 - Weekly progress reviews & plan updates

Handoff: Assignment plan to all members; used by Weekly progress reviews and by every task owner.

5.8 Review / Approval

Reviewed By: Not required

Review Responsibility: No separate reviewer; the team confirms the plan at its first review.

Task 7 - Review of real-time tracking & streaming technologies

5.1 Basic Information

Attribute	Details
Phase	Literature Review
Primary Owner	Omar Moharam
Supporting Members	None
Start Date	08/10/2026
End Date	14/10/2026
Duration	7 days
Priority	Medium
Complexity	Medium
Dependencies	Task 2 - Review scope, research questions & source-collection plan

5.2 Task Objective

Review the technologies for moving and processing location data in real time. The findings help the team choose a realistic architecture and technology stack.

5.3 Task Responsibilities

Primary Owner - Omar Moharam

- Collect sources on real-time data transport and processing options
- Compare options by latency, scalability and complexity
- Write the review section

5.4 Expected Deliverables

- Review section with an option comparison table

5.5 Definition of Done

- Main technology options are covered
- Comparison criteria are the same for every option
- Sources are cited

5.6 Key Quality Checks

- ☐ Comparison considers real-time latency needs
- ☐ Trade-offs are explained, not only listed
- ☐ Terminology matches the other review sections

5.7 Handoff / Dependency

Output Used By: Task 16 - Architecture analysis & candidate evaluation; Task 18 - Comparative synthesis & gap analysis

Handoff: Review section to Mohamed (Comparative synthesis), Mina (Architecture analysis) and Mustafa (Technology stack).

5.8 Review / Approval

Reviewed By: Mohamed Saber

Review Responsibility: Check clarity, consistent criteria and citations.

Task 8 - User stories

5.1 Basic Information

Attribute	Details
Phase	Requirements Gathering
Primary Owner	Ammar Amr
Supporting Members	None
Start Date	08/10/2026
End Date	12/10/2026
Duration	5 days
Priority	High
Complexity	Medium
Dependencies	Task 3 - Stakeholder analysis

5.2 Task Objective

Express stakeholder needs as short user stories with acceptance criteria. They connect the stakeholder analysis to the use cases and requirements.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Write user stories per stakeholder in the agreed format
- Add acceptance criteria to every story
- Prioritize the stories with the Team Leader

5.4 Expected Deliverables

- User story list with priority and acceptance criteria

5.5 Definition of Done

- Every stakeholder need has at least one story
- Every story has acceptance criteria
- Priorities are set

5.6 Key Quality Checks

- ☐ Stories are testable
- ☐ No duplicated stories
- ☐ Wording follows one format

5.7 Handoff / Dependency

Output Used By: Task 11 - Functional requirements; Task 12 - Use cases; Task 14 - Non-functional requirements; Task 22 - Requirements validation / review; Task 27 - UI/UX guidelines

Handoff: User stories to Omar (Use cases), Ammar (Functional requirements) and Mina (Non-functional requirements).

5.8 Review / Approval

Reviewed By: Omar Moharam

Review Responsibility: Check that stories are clear and cover the stakeholder needs.

Task 9 - Problem statement, objectives & scope refinement

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mustafa Mahmoud
Supporting Members	None
Start Date	08/10/2026
End Date	11/10/2026
Duration	4 days
Priority	High
Complexity	Medium
Dependencies	Task 1 - Project proposal consolidation

5.2 Task Objective

Refine the problem statement, measurable objectives and the in-scope and out-of-scope items. This sets the boundary used by analysis and design.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Refine the problem statement from the proposal
- Define measurable objectives
- Fix in-scope and out-of-scope items

5.4 Expected Deliverables

- Refined problem statement, objectives and scope statement

5.5 Definition of Done

- Approved by the team
- Out-of-scope items are explicit
- Objectives are measurable

5.6 Key Quality Checks

- ☐ Consistent with the consolidated proposal
- ☐ Boundary between design and implementation is clear
- ☐ Usable as input to the DFD and analysis

5.7 Handoff / Dependency

Output Used By: Task 28 - Context-level DFD

Handoff: Scope statement to Mina (Requirements analysis) and Mohamed (Context-level DFD).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check clarity and consistency with the proposal.

Task 10 - Risk assessment & mitigation plan

5.1 Basic Information

Attribute	Details
Phase	Project Planning & Management
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	09/10/2026
End Date	14/10/2026
Duration	6 days
Priority	Medium
Complexity	Medium
Dependencies	Task 1 - Project proposal consolidation

5.2 Task Objective

Identify the main risks to the project (schedule, technical, scope, team) and agree how to reduce them. It lets the team react early instead of finding problems late. It is one of the planning documents.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Collect risks from all members and from the Gantt dependencies
- Rate each risk by likelihood and impact and assign an owner
- Define a mitigation action for each major risk

5.4 Expected Deliverables

- Risk register: risk, likelihood, impact, owner, mitigation and trigger

5.5 Definition of Done

- Risks cover schedule, technical, scope and team areas
- Every high risk has an owner and a mitigation
- The register was discussed with the team

5.6 Key Quality Checks

- ☐ Overlap and dependency risks from the Gantt are included
- ☐ Mitigations are practical for a university team
- ☐ Risks are written clearly, not vaguely

5.7 Handoff / Dependency

Output Used By: Task 48 - Planning documents finalization

Handoff: Risk register to Mustafa (weekly reviews) and Ammar (Planning documents finalization).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check that risks are realistic and each mitigation is actionable.

Task 11 - Functional requirements

5.1 Basic Information

Attribute	Details
Phase	Requirements Gathering
Primary Owner	Ammar Amr
Supporting Members	None
Start Date	09/10/2026
End Date	15/10/2026
Duration	7 days
Priority	Critical
Complexity	High
Dependencies	Task 8 - User stories; Task 12 - Use cases

5.2 Task Objective

Define what the system must do as numbered, clear requirements. They are the main input to analysis, design and testing.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Convert stories and use cases into numbered functional requirements
- Group requirements by feature area
- Record priority and source for each requirement

5.4 Expected Deliverables

- Functional requirements list with ID, description, priority and source

5.5 Definition of Done

- Every requirement is unambiguous and testable
- Every requirement traces to a story or use case
- Requirements are grouped and prioritized

5.6 Key Quality Checks

- ☐ No duplicate or conflicting requirements
- ☐ Attendance, tracking and anti-tampering functions are covered
- ☐ IDs are consistent

5.7 Handoff / Dependency

Output Used By: Task 21 - Requirements analysis; Task 22 - Requirements validation / review; Task 26 - ERD / conceptual data model

Handoff: Functional requirements to Mina (Requirements analysis), Mustafa (Requirements validation) and Ammar (Baseline SRS).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check clarity, testability and coverage.

Task 12 - Use cases

5.1 Basic Information

Attribute	Details
Phase	Requirements Gathering
Primary Owner	Omar Moharam
Supporting Members	None
Start Date	10/10/2026
End Date	14/10/2026
Duration	5 days
Priority	High
Complexity	Medium
Dependencies	Task 3 - Stakeholder analysis; Task 8 - User stories

5.2 Task Objective

Identify what each actor wants to achieve with the system. The use cases structure the requirements and later become diagrams and descriptions.

5.3 Task Responsibilities

Primary Owner - Omar Moharam

- Derive use cases from stakeholders and user stories
- Name actors and goals
- Group and prioritize the use cases

5.4 Expected Deliverables

- Use case list with actors, goals and priority

5.5 Definition of Done

- All stakeholder goals are covered
- Each use case traces to at least one user story
- The list is approved by the team

5.6 Key Quality Checks

- ☐ Use case names are consistent
- ☐ Actors are correct
- ☐ There is no overlap between use cases

5.7 Handoff / Dependency

Output Used By: Task 11 - Functional requirements; Task 17 - Use case diagram; Task 20 - Use case descriptions; Task 22 - Requirements validation / review

Handoff: Use case list to Omar (Use case diagram and descriptions) and Ammar (Functional requirements).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check completeness and traceability to the user stories.

Task 13 - Weekly progress reviews & plan updates

5.1 Basic Information

Attribute	Details
Phase	Project Planning & Management
Primary Owner	Mustafa Mahmoud
Supporting Members	None
Start Date	11/10/2026
End Date	29/10/2026
Duration	19 days
Priority	High
Complexity	Low
Dependencies	Task 6 - Task Assignment & Roles Planning

5.2 Task Objective

Hold a regular review of progress against the Gantt Chart and update the plan when needed. It exposes delays and blockers early. It runs through the whole phase.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Run the weekly review and record the status of each active task
- Identify blockers and decide actions
- Update the plan and communicate any change

5.4 Expected Deliverables

- Weekly status note: progress, blockers, decisions and next actions

5.5 Definition of Done

- A review was held in each week between 11/10 and 29/10
- Every blocker has an owner and a date
- Plan changes are recorded and shared

5.6 Key Quality Checks

- ☐ Critical-path tasks (architecture and data model) are checked every week
- ☐ Delays reported by owners are acted on
- ☐ Notes are shared with all members

5.7 Handoff / Dependency

Output Used By: Task 48 - Planning documents finalization

Handoff: Status notes to all members; used as input to the Final system design review.

5.8 Review / Approval

Reviewed By: Not required

Review Responsibility: No separate reviewer; members confirm status in the meeting.

Task 14 - Non-functional requirements

5.1 Basic Information

Attribute	Details
Phase	Requirements Gathering
Primary Owner	Mina Safwat Fawzy
Supporting Members	None
Start Date	11/10/2026
End Date	16/10/2026
Duration	6 days
Priority	Critical
Complexity	High
Dependencies	Task 3 - Stakeholder analysis; Task 8 - User stories; Task 5 - Review of anti-tampering / anti-fraud mechanisms

5.2 Task Objective

Define the quality expectations of the system: performance, accuracy, security, reliability, usability and privacy. They strongly drive the architecture and technology choices.

5.3 Task Responsibilities

Primary Owner - Mina Safwat Fawzy

- Define quality requirements and make each one measurable
- Cover real-time performance and anti-tampering needs using the review findings
- Set priorities

5.4 Expected Deliverables

- NFR list with measurable targets and priorities

5.5 Definition of Done

- Every NFR has a measurable criterion
- Real-time and anti-tampering needs are covered
- Priorities are set

5.6 Key Quality Checks

- ☐ No vague statements
- ☐ Targets are realistic
- ☐ NFRs do not contradict the functional requirements

5.7 Handoff / Dependency

Output Used By: Task 16 - Architecture analysis & candidate evaluation; Task 21 - Requirements analysis; Task 22 - Requirements validation / review; Task 23 - Technology stack evaluation & selection; Task 25 - Architecture selection & high-level architecture

Handoff: NFRs to Mustafa (Architecture selection, Technology stack), Mina (Architecture analysis) and Ammar (Baseline SRS).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check measurability and security and performance coverage.

Task 15 - KPIs / project success metrics

5.1 Basic Information

Attribute	Details
Phase	Project Planning & Management
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	12/10/2026
End Date	16/10/2026
Duration	5 days
Priority	Medium
Complexity	Low
Dependencies	Task 1 - Project proposal consolidation

5.2 Task Objective

Define how the team will measure whether the project is successful. Measurable targets make later evaluation objective. The KPIs link the project objectives to concrete results.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Propose measurable success criteria linked to the objectives
- Define how and when each KPI is measured
- Agree targets with the team

5.4 Expected Deliverables

- KPI list: name, definition, target and measurement method

5.5 Definition of Done

- Each KPI is measurable
- Every project objective has at least one KPI
- The team has agreed on the targets

5.6 Key Quality Checks

- ☐ No KPI is vague (for example 'good performance')
- ☐ KPIs distinguish the pre-implementation phase from later phases
- ☐ Targets are consistent with the NFRs where relevant

5.7 Handoff / Dependency

Output Used By: Task 48 - Planning documents finalization

Handoff: KPI list to Ammar (Planning documents finalization).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check that KPIs are measurable and linked to the objectives.

Task 16 - Architecture analysis & candidate evaluation

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mina Safwat Fawzy
Supporting Members	Mustafa Mahmoud, Omar Moharam
Start Date	12/10/2026
End Date	16/10/2026
Duration	5 days
Priority	Critical
Complexity	High
Dependencies	Task 14 - Non-functional requirements; Task 4 - Review of UWB/RTLS workforce tracking and smart attendance systems; Task 5 - Review of anti-tampering / anti-fraud mechanisms; Task 7 - Review of real-time tracking & streaming technologies

5.2 Task Objective

Identify realistic architecture options and evaluate each against the requirements and constraints. Mustafa uses the result to make the architecture decision.

5.3 Task Responsibilities

Primary Owner - Mina Safwat Fawzy

- Define 2 or 3 realistic architecture candidates
- Evaluate them against NFRs and constraints
- Write the recommendation

Supporting Member - Mustafa Mahmoud

- Set evaluation criteria and priorities
- Challenge assumptions and review the recommendation

Supporting Member - Omar Moharam

- Provide real-time and streaming inputs from the technology review
- Check candidates for real-time feasibility

5.4 Expected Deliverables

- Candidate comparison matrix with recommendation

5.5 Definition of Done

- Candidates are compared on the same criteria
- A recommendation is stated with reasons
- The matrix is decision-ready

5.6 Key Quality Checks

- ☐ Real-time and anti-tampering needs are considered
- ☐ Assumptions are written down
- ☐ No candidate is excluded without a reason

5.7 Handoff / Dependency

Output Used By: Task 23 - Technology stack evaluation & selection; Task 25 - Architecture selection & high-level architecture

Handoff: Matrix to Mustafa (Architecture selection) and Mina (Technology stack input).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check completeness of the evaluation and soundness of the recommendation.

Task 17 - Use case diagram

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Omar Moharam
Supporting Members	None
Start Date	13/10/2026
End Date	16/10/2026
Duration	4 days
Priority	High
Complexity	Medium
Dependencies	Task 12 - Use cases

5.2 Task Objective

Model the actors and use cases visually. It shows the system boundary and what each actor can do.

5.3 Task Responsibilities

Primary Owner - Omar Moharam

- Model actors and use cases
- Show include and extend relationships correctly
- Define the system boundary

5.4 Expected Deliverables

- Use case diagram(s)

5.5 Definition of Done

- All use cases are shown
- Actors and system boundary are correct
- The diagram matches the use case list

5.6 Key Quality Checks

- ☐ Names match the use case list
- ☐ Relationships are correct and not overused
- ☐ The diagram is readable

5.7 Handoff / Dependency

Output Used By: Task 20 - Use case descriptions; Task 33 - Requirements consistency review

Handoff: Diagram to Omar (Use case descriptions) and Ammar (Activity & state diagrams).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check consistency with the use case list.

Task 18 - Comparative synthesis & gap analysis

5.1 Basic Information

Attribute	Details
Phase	Literature Review
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	14/10/2026
End Date	17/10/2026
Duration	4 days
Priority	Medium
Complexity	Medium
Dependencies	Task 4 - Review of UWB/RTLS workforce tracking and smart attendance systems; Task 5 - Review of anti-tampering / anti-fraud mechanisms; Task 7 - Review of real-time tracking & streaming technologies

5.2 Task Objective

Combine the three reviews into one comparison and identify what existing solutions do not solve. The gaps justify the project and guide its requirements.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Compare findings of the three reviews on common criteria
- List gaps and unmet needs
- Link each gap to a project objective

5.4 Expected Deliverables

- Comparison table and gap list linked to the project objectives

5.5 Definition of Done

- All three reviews are reflected
- Each gap is clear and supported by sources
- Each gap is linked to an objective

5.6 Key Quality Checks

- ☐ No gap is claimed without evidence
- ☐ Comparison criteria are consistent
- ☐ The result is understandable without reading every source

5.7 Handoff / Dependency

Output Used By: Task 19 - Literature review document drafting

Handoff: Gap analysis to Mohamed (Literature review drafting) and Mustafa (as justification for scope).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check that the gaps are supported by the reviews.

Task 19 - Literature review document drafting

5.1 Basic Information

Attribute	Details
Phase	Literature Review
Primary Owner	Mohamed Saber
Supporting Members	Mina Safwat Fawzy, Omar Moharam
Start Date	15/10/2026
End Date	19/10/2026
Duration	5 days
Priority	Medium
Complexity	Medium
Dependencies	Task 18 - Comparative synthesis & gap analysis

5.2 Task Objective

Assemble the individual review sections and the synthesis into one complete document. This is the formal literature review that will be evaluated.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Define the structure and write the introduction, UWB section, synthesis and conclusion
- Merge the other authors' sections and unify style and references
- Produce a complete draft ready for feedback

Supporting Member - Mina Safwat Fawzy

- Finalize and edit the anti-tampering section for the document

Supporting Member - Omar Moharam

- Finalize and edit the streaming technologies section for the document

5.4 Expected Deliverables

- Complete draft of the literature review including references

5.5 Definition of Done

- All sections are present
- One consistent style and reference format is used
- The draft is ready for supervisor feedback

5.6 Key Quality Checks

- ☐ Structure follows the review plan
- ☐ No missing citations
- ☐ The gap analysis is included
- ☐ Terminology is consistent

5.7 Handoff / Dependency

Output Used By: Task 32 - Feedback & evaluation

Handoff: Draft to Mohamed (Feedback & evaluation) and to the supervisor.

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check completeness, clarity and consistency of the draft with the review plan.

Task 20 - Use case descriptions

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Omar Moharam
Supporting Members	None
Start Date	15/10/2026
End Date	19/10/2026
Duration	5 days
Priority	High
Complexity	Medium
Dependencies	Task 12 - Use cases; Task 17 - Use case diagram

5.2 Task Objective

Describe each use case in detail: actors, flows, alternatives and exceptions. The descriptions are the main source for sequence diagrams, wireframes and tests.

5.3 Task Responsibilities

Primary Owner - Omar Moharam

- Write structured descriptions: preconditions, main flow, alternative flows, postconditions
- Link each description to functional requirement IDs
- Describe exceptions such as tampering alerts

5.4 Expected Deliverables

- Use case description set

5.5 Definition of Done

- Every use case is fully described
- Requirement links are added
- Descriptions were reviewed

5.6 Key Quality Checks

- ☐ Flows are complete and in order
- ☐ Exceptions are covered
- ☐ Consistent with the use case diagram

5.7 Handoff / Dependency

Output Used By: Task 29 - Wireframes; Task 33 - Requirements consistency review; Task 35 - Sequence diagrams; Task 37 - Activity & state diagrams; Task 44 - Testing & validation planning

Handoff: Descriptions to Omar (Sequence diagrams, Wireframes), Ammar (Activity & state diagrams, Testing planning) and Mina (Requirements analysis).

5.8 Review / Approval

Reviewed By: Mohamed Saber

Review Responsibility: Check completeness and consistency with the diagram and requirements.

Task 21 - Requirements analysis

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mina Safwat Fawzy
Supporting Members	None
Start Date	15/10/2026
End Date	19/10/2026
Duration	5 days
Priority	Critical
Complexity	High
Dependencies	Task 11 - Functional requirements; Task 14 - Non-functional requirements

5.2 Task Objective

Analyze the requirements for gaps, conflicts and dependencies and prepare them for design. It is the bridge between requirements and design models.

5.3 Task Responsibilities

Primary Owner - Mina Safwat Fawzy

- Analyze requirements for gaps, conflicts and dependencies
- Classify and prioritize requirements
- Identify the data and process models needed

5.4 Expected Deliverables

- Requirements analysis report and priority matrix

5.5 Definition of Done

- Gaps and conflicts are listed and resolved or escalated
- Priorities are agreed
- The report can be used by data and process modeling

5.6 Key Quality Checks

- ☐ All FR and NFR are covered
- ☐ Dependencies are listed
- ☐ Open issues have owners

5.7 Handoff / Dependency

Output Used By: Task 26 - ERD / conceptual data model; Task 28 - Context-level DFD; Task 33 - Requirements consistency review; Task 37 - Activity & state diagrams

Handoff: Analysis report to Mina (ERD), Mohamed (DFDs) and Ammar (Activity & state diagrams, Class diagram).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check coverage of all requirements.

Task 22 - Requirements validation / review

5.1 Basic Information

Attribute	Details
Phase	Requirements Gathering
Primary Owner	Mustafa Mahmoud
Supporting Members	Mohamed Saber, Omar Moharam
Start Date	16/10/2026
End Date	18/10/2026
Duration	3 days
Priority	High
Complexity	Medium
Dependencies	Task 8 - User stories; Task 12 - Use cases; Task 11 - Functional requirements; Task 14 - Non-functional requirements

5.2 Task Objective

Check user stories, use cases, FR and NFR together for completeness, clarity and conflicts. This is the quality gate before the baseline SRS.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Lead the validation session and decide on conflicts
- Review NFR and security-related items
- Approve the validated set for the SRS

Supporting Member - Mohamed Saber

- Independently check requirements for clarity and testability
- Record issues in the issue log

Supporting Member - Omar Moharam

- Check requirements against use cases and stakeholders
- Record missing or conflicting items

5.4 Expected Deliverables

- Validation report with issue log and decisions

5.5 Definition of Done

- All requirement groups are reviewed
- Every issue is resolved or assigned
- The validated set is approved by the Team Leader

5.6 Key Quality Checks

- ☐ Requirements are testable and traceable
- ☐ No conflicts remain between FR and NFR
- ☐ Use cases cover all functional requirements

5.7 Handoff / Dependency

Output Used By: Task 24 - Requirements documentation (baseline SRS)

Handoff: Validated requirements to Ammar (Baseline SRS) and Mina (Requirements analysis).

5.8 Review / Approval

Reviewed By: Not required

Review Responsibility: This task is itself the review; no separate reviewer.

Task 23 - Technology stack evaluation & selection

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mustafa Mahmoud
Supporting Members	Ammar Amr
Start Date	16/10/2026
End Date	20/10/2026
Duration	5 days
Priority	Critical
Complexity	High
Dependencies	Task 16 - Architecture analysis & candidate evaluation; Task 14 - Non-functional requirements

5.2 Task Objective

Compare technologies against the requirements and the architecture direction and decide. The decision affects the database, API and deployment design.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Define evaluation criteria
- Compare candidate technologies
- Make and record the decision

Supporting Member - Ammar Amr

- Gather option data and prepare the comparison
- Check team skills and constraints

5.4 Expected Deliverables

- Evaluation matrix and selected technology stack with justification

5.5 Definition of Done

- The decision is recorded with reasons
- It is consistent with the architecture and NFRs
- Accepted by the team

5.6 Key Quality Checks

- ☐ Meets real-time and security needs
- ☐ Realistic for the team's skills
- ☐ No unresolved conflict with the architecture

5.7 Handoff / Dependency

Output Used By: Task 40 - Physical schema; Task 42 - Deployment diagram

Handoff: Selected stack to Mina (Physical schema) and Ammar (Deployment diagram).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check the decision against the NFRs.

Task 24 - Requirements documentation (baseline SRS)

5.1 Basic Information

Attribute	Details
Phase	Requirements Gathering
Primary Owner	Ammar Amr
Supporting Members	None
Start Date	17/10/2026
End Date	20/10/2026
Duration	4 days
Priority	Critical
Complexity	Medium
Dependencies	Task 22 - Requirements validation / review

5.2 Task Objective

Assemble the validated requirements into the baseline Software Requirements Specification. It is the official reference for design and later implementation.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Assemble the validated requirements in the SRS structure
- Add scope, definitions, constraints and assumptions
- Apply the validation fixes and number everything

5.4 Expected Deliverables

- Baseline SRS including all validated functional and non-functional requirements

5.5 Definition of Done

- All validated requirements are included
- The document structure is complete
- It has been reviewed by the Team Leader

5.6 Key Quality Checks

- ☐ Numbering is consistent
- ☐ Use cases are referenced
- ☐ All validation issues are applied

5.7 Handoff / Dependency

Output Used By: Task 33 - Requirements consistency review; Task 44 - Testing & validation planning; Task 49 - Documentation consolidation

Handoff: Baseline SRS to Mohamed (Requirements consistency review), Ammar (Testing planning) and Mina (Documentation consolidation).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check that the SRS is complete and consistent with the validated requirements.

Task 25 - Architecture selection & high-level architecture

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mustafa Mahmoud
Supporting Members	Mina Safwat Fawzy, Ammar Amr
Start Date	17/10/2026
End Date	20/10/2026
Duration	4 days
Priority	Critical
Complexity	High
Dependencies	Task 16 - Architecture analysis & candidate evaluation; Task 14 - Non-functional requirements

5.2 Task Objective

Select the architecture and define its high-level structure. This is the most important design decision; components, deployment and technology depend on it.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Select the architecture and record the reasons
- Define layers and main structure
- Present the decision to the team

Supporting Member - Mina Safwat Fawzy

- Review the selection against NFRs, especially security and data needs

Supporting Member - Ammar Amr

- Check coverage of the functional requirements

5.4 Expected Deliverables

- High-level architecture diagram and decision record

5.5 Definition of Done

- The selected architecture is documented and accepted by the team
- The reasons for the choice are recorded
- It supports the key NFRs

5.6 Key Quality Checks

- ☐ Real-time, security and scalability needs are satisfied
- ☐ Consistent with the SRS
- ☐ Diagram is clear and complete

5.7 Handoff / Dependency

Output Used By: Task 30 - Major components & their interactions; Task 42 - Deployment diagram

Handoff: Architecture to Mustafa (Major components), Ammar (Deployment diagram) and Mina (data design).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check the architecture against the NFRs.

Task 26 - ERD / conceptual data model

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mina Safwat Fawzy
Supporting Members	Ammar Amr
Start Date	17/10/2026
End Date	20/10/2026
Duration	4 days
Priority	Critical
Complexity	High
Dependencies	Task 11 - Functional requirements; Task 21 - Requirements analysis

5.2 Task Objective

Identify the entities and relationships the system must store. It is the start of the database design chain.

5.3 Task Responsibilities

Primary Owner - Mina Safwat Fawzy

- Identify entities, attributes and relationships
- Draw the ERD
- Resolve ambiguities with the requirements

Supporting Member - Ammar Amr

- Map the functional requirements to entities and attributes
- Check that no data requirement is missing

5.4 Expected Deliverables

- Conceptual ERD with entity descriptions

5.5 Definition of Done

- Every data requirement is represented
- Relationships and cardinalities are defined
- The ERD matches the use cases

5.6 Key Quality Checks

- ☐ No missing entity from the SRS
- ☐ Names are consistent
- ☐ No unnecessary duplication

5.7 Handoff / Dependency

Output Used By: Task 31 - Detailed DFD; Task 34 - Logical schema; Task 43 - Class diagram

Handoff: ERD to Mina (Logical schema), Mohamed (Detailed DFD) and Ammar (Class diagram).

5.8 Review / Approval

Reviewed By: Omar Moharam

Review Responsibility: Check the ERD against the use cases.

Task 27 - UI/UX guidelines

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Omar Moharam
Supporting Members	None
Start Date	17/10/2026
End Date	20/10/2026
Duration	4 days
Priority	Medium
Complexity	Low
Dependencies	Task 8 - User stories

5.2 Task Objective

Define the visual and interaction rules for the user interface. Common rules keep the design consistent across screens.

5.3 Task Responsibilities

Primary Owner - Omar Moharam

- Define layout, navigation and component rules
- Define accessibility and usability principles
- Document them clearly

5.4 Expected Deliverables

- UI/UX guideline document

5.5 Definition of Done

- The guidelines are specific enough to design from
- They cover the main user roles
- Reviewed

5.6 Key Quality Checks

- ☐ Rules are consistent
- ☐ Accessibility is considered
- ☐ Matches the user stories

5.7 Handoff / Dependency

Output Used By: Task 29 - Wireframes

Handoff: Guidelines to Omar (Wireframes and Mockups).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check usability and consistency.

Task 28 - Context-level DFD

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	18/10/2026
End Date	19/10/2026
Duration	2 days
Priority	Medium
Complexity	Low
Dependencies	Task 9 - Problem statement, objectives & scope refinement; Task 21 - Requirements analysis

5.2 Task Objective

Show the system as one process with its external entities and main data flows. It confirms the system boundary.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Identify external entities and data flows
- Draw the context diagram
- Check it against the scope statement

5.4 Expected Deliverables

- Context-level DFD

5.5 Definition of Done

- All external entities and flows are included
- The boundary matches the scope
- Reviewed

5.6 Key Quality Checks

- ☐ Entities match the stakeholders and actors
- ☐ Flows are named clearly
- ☐ Consistent with the use case diagram

5.7 Handoff / Dependency

Output Used By: Task 31 - Detailed DFD

Handoff: Context DFD to Mohamed (Detailed DFD).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check consistency with scope and requirements.

Task 29 - Wireframes

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Omar Moharam
Supporting Members	None
Start Date	18/10/2026
End Date	22/10/2026
Duration	5 days
Priority	Medium
Complexity	Medium
Dependencies	Task 27 - UI/UX guidelines; Task 20 - Use case descriptions

5.2 Task Objective

Create low-fidelity layouts for the key screens. They let the team check the user flow before detailed design.

5.3 Task Responsibilities

Primary Owner - Omar Moharam

- Identify the key screens from use cases
- Create low-fidelity layouts
- Show navigation between screens

5.4 Expected Deliverables

- Wireframe set

5.5 Definition of Done

- Every key use case has a screen
- Navigation is shown
- Reviewed

5.6 Key Quality Checks

- ☐ Follows the guidelines
- ☐ Screens match the requirements
- ☐ Layouts are clear

5.7 Handoff / Dependency

Output Used By: Task 41 - Mockups / interactive prototype

Handoff: Wireframes to Omar (Mockups).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check coverage of the use cases.

Task 30 - Major components & their interactions

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mustafa Mahmoud
Supporting Members	Omar Moharam
Start Date	19/10/2026
End Date	22/10/2026
Duration	4 days
Priority	Critical
Complexity	High
Dependencies	Task 25 - Architecture selection & high-level architecture

5.2 Task Objective

Break the architecture into its main components and define how they interact. It is the base for the component diagram, sequence diagrams and integration work.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Break the architecture into components
- Define each component's responsibility and interfaces
- Describe the main interactions

Supporting Member - Omar Moharam

- Check components against use cases and flows
- Identify missing interactions

5.4 Expected Deliverables

- Component list with responsibilities, interfaces and interactions

5.5 Definition of Done

- Every requirement maps to a component
- No orphan components exist
- Interfaces are described

5.6 Key Quality Checks

- ☐ Consistent with the high-level architecture
- ☐ Responsibilities do not overlap
- ☐ Usable by the next tasks

5.7 Handoff / Dependency

Output Used By: Task 35 - Sequence diagrams; Task 36 - Integration considerations; Task 38 - Component diagram

Handoff: Components to Mohamed (Component diagram), Omar (Sequence diagrams) and Ammar (Integration considerations).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check consistency with the architecture and coverage of requirements.

Task 31 - Detailed DFD

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	19/10/2026
End Date	22/10/2026
Duration	4 days
Priority	Medium
Complexity	Medium
Dependencies	Task 28 - Context-level DFD; Task 26 - ERD / conceptual data model

5.2 Task Objective

Decompose the context diagram into processes, data stores and flows. It shows how data moves inside the system.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Decompose the context DFD into processes
- Define data stores and flows
- Check balance with the context DFD and ERD

5.4 Expected Deliverables

- Detailed DFD (level 1 and level 2 where needed)

5.5 Definition of Done

- Balanced with the context DFD
- Data stores match the ERD
- All main processes are covered

5.6 Key Quality Checks

- ☐ Process names are consistent
- ☐ No process without input or output
- ☐ Consistent with the requirements

5.7 Handoff / Dependency

Output Used By: Task 50 - Cross-deliverable integration review & corrections

Handoff: Detailed DFD to Mustafa (Cross-deliverable integration review).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check coverage of requirements.

Task 32 - Feedback & evaluation

5.1 Basic Information

Attribute	Details
Phase	Literature Review
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	20/10/2026
End Date	22/10/2026
Duration	3 days
Priority	Low
Complexity	Low
Dependencies	Task 19 - Literature review document drafting

5.2 Task Objective

Collect feedback on the draft from the supervisor and team and evaluate it. It shows what must be improved before the final version.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Send the draft to the supervisor and team
- Collect and record all feedback
- Prioritize feedback and assign changes

5.4 Expected Deliverables

- Feedback log with a decision for each item

5.5 Definition of Done

- Feedback is received from the supervisor and at least two team members
- Every item has a decision (accept, reject or discuss)
- Accepted items are ready for revision

5.6 Key Quality Checks

- ☐ No feedback is lost
- ☐ Rejected items have a reason
- ☐ Deadlines for revisions are clear

5.7 Handoff / Dependency

Output Used By: Task 39 - Suggested improvements & revisions

Handoff: Feedback log to Mohamed (Suggested improvements & revisions).

5.8 Review / Approval

Reviewed By: Not required

Review Responsibility: No separate reviewer; the log is shared with the team.

Task 33 - Requirements consistency review

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	20/10/2026
End Date	21/10/2026
Duration	2 days
Priority	High
Complexity	Medium
Dependencies	Task 24 - Requirements documentation (baseline SRS); Task 17 - Use case diagram; Task 20 - Use case descriptions; Task 21 - Requirements analysis

5.2 Task Objective

Independently check that the SRS, use cases, use case diagram and analysis agree with each other. It finds contradictions before design goes further.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Cross-check the SRS, use cases, use case diagram and analysis
- Record inconsistencies in a log
- Verify the fixes with the owners

5.4 Expected Deliverables

- Consistency review log with owners and decisions

5.5 Definition of Done

- All four artifacts are cross-checked
- Every inconsistency has an owner and decision
- The log is shared with the team

5.6 Key Quality Checks

- ☐ Names, IDs and actors are consistent
- ☐ No requirement lacks a use case and no use case lacks a requirement
- ☐ Fixes are confirmed

5.7 Handoff / Dependency

Output Used By: Task 50 - Cross-deliverable integration review & corrections

Handoff: Log to the artifact owners and to Mustafa (Cross-deliverable integration review).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check that the review covered all artifacts.

Task 34 - Logical schema

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mina Safwat Fawzy
Supporting Members	None
Start Date	20/10/2026
End Date	23/10/2026
Duration	4 days
Priority	Critical
Complexity	High
Dependencies	Task 26 - ERD / conceptual data model

5.2 Task Objective

Convert the conceptual model into relations, keys and constraints. It defines the data structure independent of the chosen database product.

5.3 Task Responsibilities

Primary Owner - Mina Safwat Fawzy

- Convert the ERD into relations, keys and constraints
- Normalize the design
- Document the business rules

5.4 Expected Deliverables

- Logical schema document

5.5 Definition of Done

- Every entity from the ERD is covered
- Normalization is applied
- Business rules are documented

5.6 Key Quality Checks

- ☐ Consistent with the ERD
- ☐ Keys and relationships are correct
- ☐ Naming is consistent

5.7 Handoff / Dependency

Output Used By: Task 40 - Physical schema; Task 43 - Class diagram; Task 45 - API documentation; Task 50 - Cross-deliverable integration review & corrections

Handoff: Logical schema to Mina (Physical schema and API documentation) and Ammar (Class diagram).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check consistency with the ERD and quality of the design.

Task 35 - Sequence diagrams

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Omar Moharam
Supporting Members	None
Start Date	20/10/2026
End Date	24/10/2026
Duration	5 days
Priority	High
Complexity	Medium
Dependencies	Task 20 - Use case descriptions; Task 30 - Major components & their interactions

5.2 Task Objective

Show how components and actors interact over time for the main use cases. They give the detailed picture for API and integration design.

5.3 Task Responsibilities

Primary Owner - Omar Moharam

- Select the main use cases to model
- Draw the interactions between actors and components
- Include alternative and error flows where important

5.4 Expected Deliverables

- Sequence diagram set

5.5 Definition of Done

- Every main use case is covered
- Components match the component design
- Diagrams were reviewed

5.6 Key Quality Checks

- ☐ Components and messages are named as in the design
- ☐ Main and alternative flows are shown
- ☐ Consistent with the use case descriptions

5.7 Handoff / Dependency

Output Used By: Task 45 - API documentation; Task 50 - Cross-deliverable integration review & corrections

Handoff: Sequence diagrams to Mina (API documentation) and Mohamed (Component diagram).

5.8 Review / Approval

Reviewed By: Ammar Amr

Review Responsibility: Check consistency with the use cases and components.

Task 36 - Integration considerations

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Ammar Amr
Supporting Members	None
Start Date	20/10/2026
End Date	23/10/2026
Duration	4 days
Priority	Medium
Complexity	Medium
Dependencies	Task 30 - Major components & their interactions

5.2 Task Objective

Identify how components and external systems exchange data and what risks exist. It prepares the integration work for implementation.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Identify integration points
- Describe data exchange and approach
- List risks and dependencies

5.4 Expected Deliverables

- Integration considerations document

5.5 Definition of Done

- Each interface has an integration approach
- Risks are listed
- Consistent with the component design

5.6 Key Quality Checks

- ☐ All interfaces are covered
- ☐ Data exchange is described
- ☐ Consistent with the API documentation

5.7 Handoff / Dependency

Output Used By: Task 45 - API documentation; Task 47 - Deployment strategy; Task 50 - Cross-deliverable integration review & corrections

Handoff: Document to Mina (API documentation) and Ammar (Deployment strategy).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check consistency with the components.

Task 37 - Activity & state diagrams

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Ammar Amr
Supporting Members	None
Start Date	21/10/2026
End Date	24/10/2026
Duration	4 days
Priority	Medium
Complexity	Medium
Dependencies	Task 20 - Use case descriptions; Task 21 - Requirements analysis

5.2 Task Objective

Model the main workflows and the states of important entities. They clarify the system behavior.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Model the main workflows as activity diagrams
- Model states of key entities
- Check against use case descriptions

5.4 Expected Deliverables

- Activity and state diagrams

5.5 Definition of Done

- Main workflows and entity states are covered
- Consistent with the use cases
- Reviewed

5.6 Key Quality Checks

- ☐ Start and end points are clear
- ☐ All states have valid transitions
- ☐ Names match the requirements

5.7 Handoff / Dependency

Output Used By: Task 50 - Cross-deliverable integration review & corrections

Handoff: Diagrams to Ammar (Class diagram) and Mustafa (Integration review).

5.8 Review / Approval

Reviewed By: Omar Moharam

Review Responsibility: Check consistency with use cases.

Task 38 - Component diagram

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	21/10/2026
End Date	24/10/2026
Duration	4 days
Priority	Medium
Complexity	Medium
Dependencies	Task 30 - Major components & their interactions

5.2 Task Objective

Draw the components and their interfaces. It gives a visual version of the component design.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Draw components and interfaces
- Show dependencies
- Check against the component list

5.4 Expected Deliverables

- Component diagram

5.5 Definition of Done

- Matches the component list
- Interfaces are shown
- Reviewed

5.6 Key Quality Checks

- ☐ Names are consistent
- ☐ Dependencies are correct
- ☐ Readable

5.7 Handoff / Dependency

Output Used By: Task 45 - API documentation; Task 50 - Cross-deliverable integration review & corrections

Handoff: Diagram to Mina (API documentation) and Ammar (Deployment diagram).

5.8 Review / Approval

Reviewed By: Omar Moharam

Review Responsibility: Check consistency with the design.

Task 39 - Suggested improvements & revisions

5.1 Basic Information

Attribute	Details
Phase	Literature Review
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	22/10/2026
End Date	24/10/2026
Duration	3 days
Priority	Low
Complexity	Low
Dependencies	Task 32 - Feedback & evaluation

5.2 Task Objective

Apply the accepted feedback to the literature review document. This produces the version that goes to the final review.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Apply all accepted feedback to the document
- Update references and structure where needed
- Prepare a short list of the changes made

5.4 Expected Deliverables

- Revised literature review and change list

5.5 Definition of Done

- All accepted items are applied
- The revised document is consistent
- The change list is available

5.6 Key Quality Checks

- ☐ No new inconsistencies were introduced
- ☐ References are still complete
- ☐ Changes match the feedback log

5.7 Handoff / Dependency

Output Used By: Task 46 - Final review against grading/evaluation criteria

Handoff: Revised document to Mohamed (Final review against grading criteria).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check that accepted feedback was applied correctly.

Task 40 - Physical schema

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mina Safwat Fawzy
Supporting Members	None
Start Date	22/10/2026
End Date	25/10/2026
Duration	4 days
Priority	High
Complexity	High
Dependencies	Task 34 - Logical schema; Task 23 - Technology stack evaluation & selection

5.2 Task Objective

Define the physical database structure for the selected technology. It prepares the data layer for implementation.

5.3 Task Responsibilities

Primary Owner - Mina Safwat Fawzy

- Define table structure, data types and indexes
- Consider the expected real-time data volume
- Align with the selected technology stack

5.4 Expected Deliverables

- Physical schema document

5.5 Definition of Done

- Every logical table is covered
- Consistent with the selected technology
- Reviewed by the Team Leader

5.6 Key Quality Checks

- ☐ Data types and indexes are justified
- ☐ Volume and performance needs are considered
- ☐ Matches the logical schema

5.7 Handoff / Dependency

Output Used By: Task 49 - Documentation consolidation; Task 50 - Cross-deliverable integration review & corrections

Handoff: Physical schema to Mina (API documentation and Documentation consolidation) and to Mustafa (Cross-deliverable integration review).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check consistency with the logical schema and tech stack.

Task 41 - Mockups / interactive prototype

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Omar Moharam
Supporting Members	Ammar Amr
Start Date	22/10/2026
End Date	27/10/2026
Duration	6 days
Priority	Medium
Complexity	High
Dependencies	Task 29 - Wireframes

5.2 Task Objective

Turn the wireframes into visual mockups and a clickable prototype. It lets stakeholders walk through the key flows.

5.3 Task Responsibilities

Primary Owner - Omar Moharam

- Design visual mockups from the wireframes
- Build a clickable prototype of the key flows
- Fix issues found in walkthroughs

Supporting Member - Ammar Amr

- Walk through the flows against the functional requirements
- Report gaps to Omar

5.4 Expected Deliverables

- Mockups and interactive prototype

5.5 Definition of Done

- Key flows can be walked through
- Matches the wireframes and guidelines
- Reviewed

5.6 Key Quality Checks

- [] Screens match the requirements
- [] Navigation works
- [] Visual style is consistent

5.7 Handoff / Dependency

Output Used By: Task 49 - Documentation consolidation; Task 50 - Cross-deliverable integration review & corrections

Handoff: Mockups to Mina (Documentation consolidation) and Mustafa (Integration review).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check the prototype against the requirements.

Task 42 - Deployment diagram

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Ammar Amr
Supporting Members	None
Start Date	22/10/2026
End Date	25/10/2026
Duration	4 days
Priority	Medium
Complexity	Medium
Dependencies	Task 25 - Architecture selection & high-level architecture; Task 23 - Technology stack evaluation & selection

5.2 Task Objective

Show how the system parts are placed on physical or virtual nodes. It supports the deployment strategy.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Identify nodes and connections
- Place components on nodes
- Align with the technology stack

5.4 Expected Deliverables

- Deployment diagram

5.5 Definition of Done

- Matches architecture and stack
- All main nodes are shown
- Reviewed

5.6 Key Quality Checks

- [] Consistent with the component diagram
- [] Connections are labeled
- [] Realistic for the project

5.7 Handoff / Dependency

Output Used By: Task 47 - Deployment strategy; Task 50 - Cross-deliverable integration review & corrections

Handoff: Diagram to Ammar (Deployment strategy).

5.8 Review / Approval

Reviewed By: Mohamed Saber

Review Responsibility: Check consistency with the architecture.

Task 43 - Class diagram

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Ammar Amr
Supporting Members	None
Start Date	23/10/2026
End Date	26/10/2026
Duration	4 days
Priority	Medium
Complexity	Medium
Dependencies	Task 26 - ERD / conceptual data model; Task 34 - Logical schema

5.2 Task Objective

Model the main classes, attributes and relationships. It connects the data model to the future implementation structure.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Define the main classes and attributes
- Define relationships and key operations
- Align with the ERD and logical schema

5.4 Expected Deliverables

- Class diagram

5.5 Definition of Done

- Consistent with the ERD and schema
- All main concepts are represented
- Reviewed

5.6 Key Quality Checks

- ☐ Class names match the other diagrams
- ☐ Relationships and multiplicities are correct
- ☐ No unnecessary detail

5.7 Handoff / Dependency

Output Used By: Task 49 - Documentation consolidation; Task 50 - Cross-deliverable integration review & corrections

Handoff: Class diagram to Mustafa (Cross-deliverable integration review) and Mina (Documentation consolidation).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check consistency with the data model.

Task 44 - Testing & validation planning

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Ammar Amr
Supporting Members	Mohamed Saber
Start Date	23/10/2026
End Date	27/10/2026
Duration	5 days
Priority	Medium
Complexity	Medium
Dependencies	Task 24 - Requirements documentation (baseline SRS); Task 20 - Use case descriptions

5.2 Task Objective

Plan how the system will be tested and validated against the requirements. It makes later testing organized and traceable.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Define test levels and approach
- Define traceability to requirements
- Define acceptance criteria

Supporting Member - Mohamed Saber

- Draft the traceability table from the requirements
- Prepare test scenarios from use cases

5.4 Expected Deliverables

- Test and validation plan

5.5 Definition of Done

- Every requirement group has a test approach
- Traceable to the SRS
- Reviewed

5.6 Key Quality Checks

- ☐ Acceptance criteria are measurable
- ☐ Covers NFRs
- ☐ Realistic

5.7 Handoff / Dependency

Output Used By: Task 49 - Documentation consolidation; Task 50 - Cross-deliverable integration review & corrections

Handoff: Plan to Mina (Documentation consolidation).

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Check traceability.

Task 45 - API documentation

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mina Safwat Fawzy
Supporting Members	Omar Moharam
Start Date	24/10/2026
End Date	27/10/2026
Duration	4 days
Priority	High
Complexity	High
Dependencies	Task 35 - Sequence diagrams; Task 34 - Logical schema; Task 36 - Integration considerations; Task 38 - Component diagram

5.2 Task Objective

Document the interfaces between components and with clients. It lets the implementation team work independently.

5.3 Task Responsibilities

Primary Owner - Mina Safwat Fawzy

- Define endpoints, requests and responses
- Define error handling
- Check against the schema and sequence diagrams

Supporting Member - Omar Moharam

- Provide endpoints and message details from the sequence diagrams
- Check examples against the flows

5.4 Expected Deliverables

- API specification document

5.5 Definition of Done

- Main flows have documented endpoints
- Names match the schema
- Reviewed

5.6 Key Quality Checks

- ☐ Consistent with sequence diagrams
- ☐ Errors are covered
- ☐ Data fields match the schema

5.7 Handoff / Dependency

Output Used By: Task 49 - Documentation consolidation; Task 50 - Cross-deliverable integration review & corrections

Handoff: API specification to Mina (Documentation consolidation) and Mustafa (Integration review).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check completeness and consistency.

Task 46 - Final review against grading/evaluation criteria

5.1 Basic Information

Attribute	Details
Phase	Literature Review
Primary Owner	Mohamed Saber
Supporting Members	None
Start Date	25/10/2026
End Date	27/10/2026
Duration	3 days
Priority	Medium
Complexity	Low
Dependencies	Task 39 - Suggested improvements & revisions

5.2 Task Objective

Check the final literature review against the grading and evaluation criteria. This makes sure the document meets the requirements before it is consolidated.

5.3 Task Responsibilities

Primary Owner - Mohamed Saber

- Build a checklist from the grading criteria
- Check the document against every criterion
- Fix remaining gaps or report them to the Team Leader

5.4 Expected Deliverables

- Completed criteria checklist and final approved literature review

5.5 Definition of Done

- Every criterion is met or has a documented exception
- The final version is approved by the Team Leader
- The document is ready for consolidation

5.6 Key Quality Checks

- ☐ Format and length requirements are met
- ☐ Required sections are present
- ☐ Citations are complete

5.7 Handoff / Dependency

Output Used By: Task 49 - Documentation consolidation

Handoff: Final literature review to Mina (Documentation consolidation).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Give final approval of the literature review.

Task 47 - Deployment strategy

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Ammar Amr
Supporting Members	None
Start Date	25/10/2026
End Date	27/10/2026
Duration	3 days
Priority	Medium
Complexity	Low
Dependencies	Task 42 - Deployment diagram; Task 36 - Integration considerations

5.2 Task Objective

Describe how the system will be deployed and rolled out. It prepares the setup for implementation.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Describe environments
- Describe rollout and hardware setup
- List risks

5.4 Expected Deliverables

- Deployment strategy document

5.5 Definition of Done

- Consistent with the deployment diagram
- Realistic for the project
- Reviewed

5.6 Key Quality Checks

- ☐ Covers all environments
- ☐ Consistent with the stack
- ☐ Risks are listed

5.7 Handoff / Dependency

Output Used By: Task 49 - Documentation consolidation; Task 50 - Cross-deliverable integration review & corrections

Handoff: Strategy to Mina (Documentation consolidation).

5.8 Review / Approval

Reviewed By: Omar Moharam

Review Responsibility: Check consistency with the deployment diagram.

Task 48 - Planning documents finalization

5.1 Basic Information

Attribute	Details
Phase	Project Planning & Management
Primary Owner	Ammar Amr
Supporting Members	None
Start Date	26/10/2026
End Date	30/10/2026
Duration	5 days
Priority	Medium
Complexity	Low
Dependencies	Task 10 - Risk assessment & mitigation plan; Task 15 - KPIs / project success metrics; Task 13 - Weekly progress reviews & plan updates

5.2 Task Objective

Bring all planning documents (assignment plan, risk register, KPIs, schedule status) to their final state for this phase. The final package becomes part of the consolidated documentation.

5.3 Task Responsibilities

Primary Owner - Ammar Amr

- Collect the latest version of each planning document
- Update them with decisions and changes made during the phase
- Check dates, names and owners for consistency across them

5.4 Expected Deliverables

- Final planning documents: assignment plan, risk register, KPI list and final schedule status

5.5 Definition of Done

- All planning documents are in their final state
- Dates and owners match the final Gantt
- Reviewed and accepted by the Team Leader

5.6 Key Quality Checks

- ☐ Risk register is closed or updated
- ☐ KPI list matches the final scope
- ☐ Version numbers are consistent

5.7 Handoff / Dependency

Output Used By: Task 52 - Final consistency check & deliverable preparation

Handoff: Final planning documents to Mina (Documentation consolidation) and Mustafa (Final consistency check).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check completeness and alignment with the final schedule.

Task 49 - Documentation consolidation

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mina Safwat Fawzy
Supporting Members	Mohamed Saber, Ammar Amr
Start Date	26/10/2026
End Date	30/10/2026
Duration	5 days
Priority	High
Complexity	Medium
Dependencies	Task 46 - Final review against grading/evaluation criteria; Task 24 - Requirements documentation (baseline SRS); Task 40 - Physical schema; Task 43 - Class diagram; Task 41 - Mockups / interactive prototype; Task 45 - API documentation; Task 44 - Testing & validation planning; Task 47 - Deployment strategy

5.2 Task Objective

Merge all deliverables into one structured document set with consistent format and references. It becomes the final handover package.

5.3 Task Responsibilities

Primary Owner - Mina Safwat Fawzy

- Define structure and numbering
- Merge deliverables
- Check cross-references

Supporting Member - Mohamed Saber

- Format and cross-reference documents
- Check versions

Supporting Member - Ammar Amr

- Check SRS and planning sections
- Provide final versions of his deliverables

5.4 Expected Deliverables

- Consolidated document set

5.5 Definition of Done

- All deliverables are included
- Format is consistent
- Reviewed

5.6 Key Quality Checks

- ☐ No missing sections
- ☐ No old versions
- ☐ Cross-references work

5.7 Handoff / Dependency

Output Used By: Task 52 - Final consistency check & deliverable preparation

Handoff: Document set to Mustafa (Final consistency check).

5.8 Review / Approval

Reviewed By: Mustafa Mahmoud

Review Responsibility: Check completeness.

Task 50 - Cross-deliverable integration review & corrections

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mustafa Mahmoud
Supporting Members	Mina Safwat Fawzy, Omar Moharam, Ammar Amr, Mohamed Saber
Start Date	27/10/2026
End Date	29/10/2026
Duration	3 days
Priority	Critical
Complexity	High
Dependencies	Task 33 - Requirements consistency review; Task 34 - Logical schema; Task 40 - Physical schema; Task 31 - Detailed DFD; Task 35 - Sequence diagrams; Task 37 - Activity & state diagrams; Task 43 - Class diagram; Task 41 - Mockups / interactive prototype; Task 36 - Integration considerations; Task 38 - Component diagram; Task 42 - Deployment diagram; Task 45 - API documentation; Task 44 - Testing & validation planning; Task 47 - Deployment strategy

5.2 Task Objective

Check that requirements, diagrams, schema, UI and API agree with each other and fix conflicts. This ensures the design package is consistent.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Lead the review
- Decide on corrections
- Confirm that corrections are done

Supporting Member - Mina Safwat Fawzy

- Check data model, API and documentation consistency
- Fix issues in her deliverables

Supporting Member - Omar Moharam

- Check use cases, sequence diagrams, UI and prototype consistency
- Fix issues in his deliverables

Supporting Member - Ammar Amr

- Check SRS, behavior diagrams, class diagram and deployment consistency
- Fix issues in his deliverables

Supporting Member - Mohamed Saber

- Check DFDs, component diagram and literature review consistency
- Fix issues in his deliverables

5.4 Expected Deliverables

- Integration review log and confirmed corrections

5.5 Definition of Done

- All conflicts are fixed or accepted
- Corrections are confirmed
- The log is shared

5.6 Key Quality Checks

- ☐ Names, entities and flows are the same everywhere
- ☐ No requirement is without design coverage
- ☐ Corrections are verified

5.7 Handoff / Dependency

Output Used By: Task 51 - Final system design review

Handoff: Corrected deliverables to Mustafa (Final system design review) and Mina (Documentation consolidation).

5.8 Review / Approval

Reviewed By: Not required

Review Responsibility: This task is itself the review; no separate reviewer.

Task 51 - Final system design review

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mustafa Mahmoud
Supporting Members	None
Start Date	28/10/2026
End Date	29/10/2026
Duration	2 days
Priority	Critical
Complexity	Medium
Dependencies	Task 50 - Cross-deliverable integration review & corrections

5.2 Task Objective

Perform the final approval review of the complete design package. It confirms that the project is ready for implementation.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Review the complete design
- List open issues
- Approve or return

5.4 Expected Deliverables

- Final review sign-off

5.5 Definition of Done

- No open critical issues
- Design is complete enough for implementation
- Sign-off is recorded

5.6 Key Quality Checks

- ☐ All deliverables included
- ☐ Decisions are recorded
- ☐ Open issues are minor

5.7 Handoff / Dependency

Output Used By: Task 52 - Final consistency check & deliverable preparation

Handoff: Sign-off to Mustafa (Final consistency check).

5.8 Review / Approval

Reviewed By: Not required

Review Responsibility: This task is itself the review; no separate reviewer.

Task 52 - Final consistency check & deliverable preparation

5.1 Basic Information

Attribute	Details
Phase	System Analysis & Design
Primary Owner	Mustafa Mahmoud
Supporting Members	None
Start Date	30/10/2026
End Date	30/10/2026
Duration	1 day
Priority	Critical
Complexity	Medium
Dependencies	Task 51 - Final system design review; Task 49 - Documentation consolidation; Task 48 - Planning documents finalization

5.2 Task Objective

Do the last check of the packaged deliverables and prepare them for handover. It closes the pre-implementation phase.

5.3 Task Responsibilities

Primary Owner - Mustafa Mahmoud

- Check the final package
- Fix last issues
- Prepare for handover

5.4 Expected Deliverables

- Final deliverable package

5.5 Definition of Done

- Package is complete, versioned and consistent
- Nothing from the Gantt is missing
- Ready for implementation

5.6 Key Quality Checks

- ☐ All tasks have outputs
- ☐ Versions are consistent
- ☐ No open issues

5.7 Handoff / Dependency

Output Used By: None

Handoff: Package to the team and supervisor.

5.8 Review / Approval

Reviewed By: Mina Safwat Fawzy

Review Responsibility: Do a second check of completeness.

6. Shared / Collaborative Tasks

Only 13 of the 52 tasks are shared. Each has one Primary Owner who stays accountable for completion; supporting members only have the responsibilities listed.

Project proposal consolidation (Task 1)

Primary Owner: Mustafa Mahmoud

Supporting Members: Mohamed Saber

Time: 04/10/2026 to 07/10/2026

Responsibility Distribution

Member	Responsibility
Mustafa Mahmoud (Primary Owner)	- Decide the final wording of scope, objectives and approach - Resolve conflicting inputs from members - Get the team's agreement on the final text
Mohamed Saber (Supporting)	- Compile team inputs and edits into one draft - Keep version control and track open comments

Final Deliverable

Consolidated proposal (baseline v1): background, objectives, scope boundaries, high-level approach and assumptions

Definition of Done

- All five members have confirmed the content
- Scope and objectives are stated once, with no conflicting versions
- The text states that this period covers planning and design only, not implementation

Architecture analysis & candidate evaluation (Task 16)

Primary Owner: Mina Safwat Fawzy

Supporting Members: Mustafa Mahmoud, Omar Moharam

Time: 12/10/2026 to 16/10/2026

Responsibility Distribution

Member	Responsibility
Mina Safwat Fawzy (Primary Owner)	- Define 2 or 3 realistic architecture candidates - Evaluate them against NFRs and constraints - Write the recommendation
Mustafa Mahmoud (Supporting)	- Set evaluation criteria and priorities - Challenge assumptions and review the recommendation
Omar Moharam (Supporting)	- Provide real-time and streaming inputs from the technology review - Check candidates for real-time feasibility

Final Deliverable

Candidate comparison matrix with recommendation

Definition of Done

- Candidates are compared on the same criteria
- A recommendation is stated with reasons
- The matrix is decision-ready

Literature review document drafting (Task 19)

Primary Owner: Mohamed Saber

Supporting Members: Mina Safwat Fawzy, Omar Moharam

Time: 15/10/2026 to 19/10/2026

Responsibility Distribution

Member	Responsibility
Mohamed Saber (Primary Owner)	- Define the structure and write the introduction, UWB section, synthesis and conclusion - Merge the other authors' sections and unify style and references - Produce a complete draft ready for feedback
Mina Safwat Fawzy (Supporting)	- Finalize and edit the anti-tampering section for the document
Omar Moharam (Supporting)	- Finalize and edit the streaming technologies section for the document

Final Deliverable

Complete draft of the literature review including references

Definition of Done

- All sections are present
- One consistent style and reference format is used
- The draft is ready for supervisor feedback

Requirements validation / review (Task 22)

Primary Owner: Mustafa Mahmoud

Supporting Members: Mohamed Saber, Omar Moharam

Time: 16/10/2026 to 18/10/2026

Responsibility Distribution

Member	Responsibility
Mustafa Mahmoud (Primary Owner)	- Lead the validation session and decide on conflicts - Review NFR and security-related items - Approve the validated set for the SRS
Mohamed Saber (Supporting)	- Independently check requirements for clarity and testability - Record issues in the issue log
Omar Moharam (Supporting)	- Check requirements against use cases and stakeholders - Record missing or conflicting items

Final Deliverable

Validation report with issue log and decisions

Definition of Done

- All requirement groups are reviewed
- Every issue is resolved or assigned
- The validated set is approved by the Team Leader

Technology stack evaluation & selection (Task 23)

Primary Owner: Mustafa Mahmoud

Supporting Members: Ammar Amr

Time: 16/10/2026 to 20/10/2026

Responsibility Distribution

Member	Responsibility
Mustafa Mahmoud (Primary Owner)	- Define evaluation criteria - Compare candidate technologies - Make and record the decision
Ammar Amr (Supporting)	- Gather option data and prepare the comparison - Check team skills and constraints

Final Deliverable

Evaluation matrix and selected technology stack with justification

Definition of Done

- The decision is recorded with reasons
- It is consistent with the architecture and NFRs
- Accepted by the team

Architecture selection & high-level architecture (Task 25)

Primary Owner: Mustafa Mahmoud

Supporting Members: Mina Safwat Fawzy, Ammar Amr

Time: 17/10/2026 to 20/10/2026

Responsibility Distribution

Member	Responsibility
Mustafa Mahmoud (Primary Owner)	- Select the architecture and record the reasons - Define layers and main structure - Present the decision to the team
Mina Safwat Fawzy (Supporting)	- Review the selection against NFRs, especially security and data needs
Ammar Amr (Supporting)	- Check coverage of the functional requirements

Final Deliverable

High-level architecture diagram and decision record

Definition of Done

- The selected architecture is documented and accepted by the team
- The reasons for the choice are recorded
- It supports the key NFRs

ERD / conceptual data model (Task 26)

Primary Owner: Mina Safwat Fawzy

Supporting Members: Ammar Amr

Time: 17/10/2026 to 20/10/2026

Responsibility Distribution

Member	Responsibility
Mina Safwat Fawzy (Primary Owner)	- Identify entities, attributes and relationships - Draw the ERD - Resolve ambiguities with the requirements
Ammar Amr (Supporting)	- Map the functional requirements to entities and attributes - Check that no data requirement is missing

Final Deliverable

Conceptual ERD with entity descriptions

Definition of Done

- Every data requirement is represented
- Relationships and cardinalities are defined
- The ERD matches the use cases

Major components & their interactions (Task 30)

Primary Owner: Mustafa Mahmoud

Supporting Members: Omar Moharam

Time: 19/10/2026 to 22/10/2026

Responsibility Distribution

Member	Responsibility
Mustafa Mahmoud (Primary Owner)	- Break the architecture into components - Define each component's responsibility and interfaces - Describe the main interactions
Omar Moharam (Supporting)	- Check components against use cases and flows - Identify missing interactions

Final Deliverable

Component list with responsibilities, interfaces and interactions

Definition of Done

- Every requirement maps to a component
- No orphan components exist
- Interfaces are described

Mockups / interactive prototype (Task 41)

Primary Owner: Omar Moharam

Supporting Members: Ammar Amr

Time: 22/10/2026 to 27/10/2026

Responsibility Distribution

Member	Responsibility
Omar Moharam (Primary Owner)	- Design visual mockups from the wireframes - Build a clickable prototype of the key flows - Fix issues found in walkthroughs
Ammar Amr (Supporting)	- Walk through the flows against the functional requirements - Report gaps to Omar

Final Deliverable

Mockups and interactive prototype

Definition of Done

- Key flows can be walked through
- Matches the wireframes and guidelines
- Reviewed

Testing & validation planning (Task 44)

Primary Owner: Ammar Amr

Supporting Members: Mohamed Saber

Time: 23/10/2026 to 27/10/2026

Responsibility Distribution

Member	Responsibility
Ammar Amr (Primary Owner)	- Define test levels and approach - Define traceability to requirements - Define acceptance criteria
Mohamed Saber (Supporting)	- Draft the traceability table from the requirements - Prepare test scenarios from use cases

Final Deliverable

Test and validation plan

Definition of Done

- Every requirement group has a test approach
- Traceable to the SRS
- Reviewed

API documentation (Task 45)**Primary Owner:** Mina Safwat Fawzy**Supporting Members:** Omar Moharam**Time:** 24/10/2026 to 27/10/2026**Responsibility Distribution**

Member	Responsibility
Mina Safwat Fawzy (Primary Owner)	- Define endpoints, requests and responses - Define error handling - Check against the schema and sequence diagrams
Omar Moharam (Supporting)	- Provide endpoints and message details from the sequence diagrams - Check examples against the flows

Final Deliverable

API specification document

Definition of Done

- Main flows have documented endpoints
- Names match the schema
- Reviewed

Documentation consolidation (Task 49)**Primary Owner:** Mina Safwat Fawzy**Supporting Members:** Mohamed Saber, Ammar Amr**Time:** 26/10/2026 to 30/10/2026**Responsibility Distribution**

Member	Responsibility
Mina Safwat Fawzy (Primary Owner)	- Define structure and numbering - Merge deliverables - Check cross-references
Mohamed Saber (Supporting)	- Format and cross-reference documents - Check versions
Ammar Amr (Supporting)	- Check SRS and planning sections - Provide final versions of his deliverables

Final Deliverable

Consolidated document set

Definition of Done

- All deliverables are included
- Format is consistent
- Reviewed

Cross-deliverable integration review & corrections (Task 50)

Primary Owner: Mustafa Mahmoud

Supporting Members: Mina Safwat Fawzy, Omar Moharam, Ammar Amr, Mohamed Saber

Time: 27/10/2026 to 29/10/2026

Responsibility Distribution

Member	Responsibility
Mustafa Mahmoud (Primary Owner)	- Lead the review - Decide on corrections - Confirm that corrections are done
Mina Safwat Fawzy (Supporting)	- Check data model, API and documentation consistency - Fix issues in her deliverables
Omar Moharam (Supporting)	- Check use cases, sequence diagrams, UI and prototype consistency - Fix issues in his deliverables
Ammar Amr (Supporting)	- Check SRS, behavior diagrams, class diagram and deployment consistency - Fix issues in his deliverables
Mohamed Saber (Supporting)	- Check DFDs, component diagram and literature review consistency - Fix issues in his deliverables

Final Deliverable

Integration review log and confirmed corrections

Definition of Done

- All conflicts are fixed or accepted
- Corrections are confirmed
- The log is shared

7. Workload Distribution Summary

Member	Primary Tasks	Shared Tasks
Mustafa Mahmoud	11	7
Mina Safwat Fawzy	9	7
Omar Moharam	9	7
Ammar Amr	10	7
Mohamed Saber	13	6

Primary Tasks counts every task the member owns. Shared Tasks counts the shared tasks (Section 6) in which the member is Primary Owner or Supporting Member. Task counts alone do not show fairness; the balance is analyzed in Section 8.

8. Workload & Timeline Risk

8.1 Workload Balance Assessment

Workload was judged by effort, not task count. Mohamed owns the most tasks, but most are short or low-complexity (planning, review sections, DFDs, literature review follow-up). Mustafa owns 11 tasks, including most of the critical architecture and final-review work, and also leads the weekly reviews; four of his tasks are only 1 to 3 days long, which keeps his load strong but manageable. Mina owns the highest concentration of high-complexity tasks (anti-tampering review, NFR, architecture analysis, requirements analysis and the full database chain). Omar owns the use case chain and the whole UI/UX chain. Ammar owns the functional requirements and SRS chain plus the deployment, behavior and test planning work. All three intermediate members therefore carry meaningful independent deliverables. The risks below are the places where this balance is most likely to break because of overlapping dates.

8.2 Identified Risks

Risk 1

Risk: Mustafa has overlapping critical decisions between 16/10 and 22/10: Requirements validation (16/10-18/10), Technology stack evaluation & selection (16/10-20/10), Architecture selection & high-level architecture (17/10-20/10) and Major components & their interactions (19/10-22/10). He also leads weekly reviews and reviews 11 tasks of other members.

Impact: A delay in the architecture or component decision delays every task that depends on it, and the quality of decisions may drop if he is overloaded.

Recommendation: Ask Mina to deliver a decision-ready comparison on 16/10 and Ammar to prepare the technology comparison data. Hold one architecture decision meeting on 17/10. Keep reviews short (checking only the items listed in each task's review responsibility) and let peer reviewers do the first check.

Risk 2

Risk: Mina has three high-effort tasks in parallel between 11/10 and 16/10: Review of anti-tampering / anti-fraud mechanisms (to 13/10), Non-functional requirements (11/10-16/10) and Architecture analysis & candidate evaluation (12/10-16/10).

Impact: Weaker NFRs or a rushed architecture evaluation, which would affect all later design tasks.

Recommendation: Mohamed pre-collects and organizes anti-tampering sources; Ammar supplies NFR candidates from the user stories by 11/10; Mina finishes the threat/countermeasure table first because the NFRs and architecture both depend on it.

Risk 3

Risk: The database chain depends on requirements analysis: ERD / conceptual data model starts on 17/10 but Requirements analysis ends on 19/10, and Mina owns both while also starting Logical schema (20/10) and Physical schema (22/10). Physical schema also needs the Technology stack decision (ends 20/10).

Impact: ERD rework and a bottleneck in the database chain, which also feeds API documentation, the class diagram and the detailed DFD.

Recommendation: Share the entity-relevant part of the requirements analysis by 17/10, let Ammar map functional requirements to entities, freeze the ERD on 20/10, and publish the technology decision before 22/10.

Risk 4

Risk: Omar has up to four tasks at the same time between 17/10 and 24/10: Use case descriptions (to 19/10), UI/UX guidelines (17/10-20/10), Wireframes (18/10-22/10) and Sequence diagrams (20/10-24/10), followed by Mockups / interactive prototype (22/10-27/10).

Impact: Late sequence diagrams delay API documentation (starts 24/10); a late prototype delays the integration review.

Recommendation: Keep UI/UX guidelines short and finish them on 20/10; model the most important use cases first in the sequence diagrams; Ammar's walkthrough support on the prototype starts as soon as wireframes are ready.

Risk 5

Risk: Ammar has up to five tasks at the same time between 23/10 and 26/10: Integration considerations (to 23/10), Activity & state diagrams (to 24/10), Deployment diagram (to 25/10), Class diagram (23/10-26/10) and Testing & validation planning (23/10-27/10), then Deployment strategy (25/10-27/10) and Planning documents finalization (from 26/10).

Impact: Late or shallow behavior, deployment or test planning deliverables just before the integration review.

Recommendation: Work in priority order (Integration considerations, Activity & state diagrams, Deployment diagram, Class diagram, Testing & validation planning). Mohamed prepares the traceability table for Testing & validation planning so Ammar can focus on the strategy.

Risk 6

Risk: Several tasks start before their input is finished: Sequence diagrams (20/10), Integration considerations (20/10) and Component diagram (21/10) start while Major components & their interactions ends on 22/10.

Impact: Work done on incomplete input may need rework, and different owners may use different component names.

Recommendation: Mustafa publishes an interim component list on 20/10 and freezes it on 22/10; owners update their diagrams on 22/10 and 23/10.

Risk 7

Risk: Requirements consistency review (20/10-21/10, Mohamed) is only two days long and needs the baseline SRS (ends 20/10) and Requirements analysis (ends 19/10).

Impact: Mohamed waits for outputs and may not have time to complete a full cross-check.

Recommendation: Ammar and Mina hand over near-final SRS and analysis drafts on 19/10; Mohamed starts checking the finished parts immediately.

Risk 8

Risk: Mohamed has up to four tasks in parallel between 19/10 and 24/10: Detailed DFD (to 22/10), Feedback & evaluation (20/10-22/10), Requirements consistency review (20/10-21/10), Component diagram (21/10-24/10) and Suggested improvements & revisions (22/10-24/10). The Component diagram also depends on a component list that ends on 22/10.

Impact: Late DFD or component diagram; delay in literature review revisions.

Recommendation: Give priority to Requirements consistency review (2 days) and the Detailed DFD; literature review feedback and revisions are low effort and can be done in short sessions. Ask Omar to confirm the component names on 22/10.

Risk 9

Risk: Documentation consolidation starts on 26/10, but several inputs end later: Class diagram (26/10), Mockups / interactive prototype, API documentation, Testing & validation planning and Deployment strategy (all 27/10). Mina is also finishing Physical schema (25/10) and API documentation (27/10).

Impact: Document versions may become outdated after consolidation, causing rework during the integration review; Mina has heavy load on 26/10 and 27/10.

Recommendation: Consolidate on a rolling basis (finished documents first). Mohamed formats and cross-references; Ammar supplies final versions of his documents on time. Use a shared version list.

Risk 10

Risk: The final window is short and contains critical tasks: Cross-deliverable integration review & corrections (27/10-29/10), Final system design review (28/10-29/10) and Final consistency check & deliverable preparation (30/10, one day).

Impact: Any late deliverable or large correction leaves almost no time before 30/10; Mustafa owns all three final tasks.

Recommendation: Freeze deliverables at the end of 27/10, give owners 24 hours to fix reported issues, and let Mina and Ammar check the final package in parallel on 30/10.

Risk 11

Risk: Feedback & evaluation (20/10-22/10) depends on feedback from the supervisor and team, and the last two literature review tasks are only 2 and 3 days long.

Impact: Late external feedback delays the revised and final literature review.

Recommendation: Send the draft on 19/10 as soon as drafting ends and agree a feedback date in advance; start applying team feedback immediately.

9. Team Responsibility Rules

Ownership

The Primary Owner is responsible for ensuring the task is completed.

Collaboration

Supporting Members contribute to the task but do not replace the Primary Owner.

Deadlines

The End Date represents the expected completion date.

Quality

A task is not considered complete simply because work has started or a draft exists. The Definition of Done must be satisfied.

Handoff

Outputs required by later tasks must be clearly transferred to the responsible member, as described in the Handoff / Dependency section of each task.

Review

Reviewers check quality and consistency. Reviewing does not automatically mean owning the task.

Accountability

If a task is delayed or blocked, the Primary Owner must communicate the issue to the Team Leader as soon as it is known, not on the deadline.

10. Final Pre-Implementation Checklist

Project Pre-Implementation Completion Checklist

- ☐ All Requirements tasks completed (stakeholder analysis, user stories, use cases, functional and non-functional requirements, baseline SRS)
- ☐ Requirements reviewed (Requirements validation / review and Requirements consistency review)
- ☐ Literature review completed and finally reviewed against the evaluation criteria
- ☐ Analysis tasks completed (problem statement, use case diagram and descriptions, requirements analysis)
- ☐ Design tasks completed (architecture, components, technology stack, integration, API, UI/UX)
- ☐ Required diagrams completed (use case, DFD, sequence, activity and state, class, component, deployment)
- ☐ Database/System design completed (ERD, logical schema, physical schema)
- ☐ Testing and deployment planning completed
- ☐ Planning documents finalized
- ☐ Required documentation completed and consolidated
- ☐ Dependencies and handoffs completed
- ☐ Major inconsistencies resolved (Cross-deliverable integration review & corrections)
- ☐ Final pre-implementation outputs reviewed (Final system design review)
- ☐ Final consistency check completed on 30/10/2026
- ☐ Project is ready to enter the Implementation phase